Jinal Mandalia

Technical Program Manager | Device Releases | Validation & Reliability | AI/ML Evaluation

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Summary

Technical program and release lead with nearly **4 years** shipping device software across firmware, apps and cloud, plus an **M.S. CS thesis on evaluating vision-language models**. Led **22+ releases to 20,000+ vehicles** with a **team of 5** and **3 suppliers** at MG Motor, driving **120+ issues** and **8+ critical field failures** to root cause.

Core Skills

Program management: Device & software release planning, readiness gates, go/no-go & launch criteria, critical path, risk & dependency management, requirements, OEM, supplier & vendor management, executive reporting, Agile, Azure DevOps

Validation & reliability: Test plans, DFMEA, root cause analysis & corrective actions, field issue triage, ECU test benches (Vector CANoe/CANalyzer, Vehicle Spy), EMI-EMC & certification testing, CI/CD (GitHub Actions)

AI/ML data: Labeled evaluation datasets, locked test sets & cross-validation, multi-model evaluation, data quality reviews, AI-assisted workflows (Claude, Gemini API), Python, C/C++, AWS

Experience

Phase Technologies — **Rapid City, SD**

Software Developer — *May 2026 – Present*

- **Own end-to-end delivery** of Phase Connect, the company's iOS/Android app for commissioning and controlling power-electronics field devices; took it from **non-functional to release-ready** across firmware, hardware and cloud teams.
- Defined the device interface with firmware over **BLE (binary Protobuf)** and **Modbus**, and built the **MQTT cloud link**.
- Introduced an **AI-assisted workflow** for features, test cases, unit tests and review on every merge request.
- Run **bug and release tracking** and **test every release on real field devices**; defined the release process: automated versioning and release notes, merge-request review and forced app updates.

University of Nevada, Reno — **Reno, NV**

AI/ML Researcher — *Jan 2025 – Present*

- **Thesis:** edge-side token compression for medical vision-language models (**~82% fewer image tokens**, **<5% quality drop**); built **evaluation sets from labeled data** (EchoNet-Dynamic, Kvasir-Capsule) with locked test sets.
- Ran **multi-model evaluations** (3 Gemini models × 60 images × 4 strategies); failure reviews caught a model bias and a frame-selection method that dropped key diagnostic frames, which led to a **redesigned selector**.
- Trained LSTM, CNN and FNN models on labeled CAN traffic for intrusion detection (**95.1% accuracy**).

Matter Motor Works — **India**

Engineer, Infotainment & Cluster Systems — *Jan 2024 – Dec 2024*

- **Owned the OTA program definition**: wrote the complete requirements across the web portal, cloud and vehicle software and **assigned each team's responsibilities**, plus **instrument-cluster requirements** to government regulatory norms.
- Designed the **end-to-end OTA architecture and update flow** with secure authentication and Azure DevOps CI/CD from build to vehicle, and validated it on ECUs; the design was later used to **launch OTA updates in vehicles**.
- Built secure **vehicle-to-cloud provisioning** for Map and SIM; ran **DFMEA** to mitigate failure modes early.

JSW MG Motor India — **India**

Deputy Manager, Infotainment & Telematics — *Aug 2022 – Jan 2024*

- **Owned the infotainment and telematics release roadmap** for MG Astor across production and OTA releases.
- Led **22+ software releases to 20,000+ vehicles** via production and OTA; managed a **team of 5** for testing and flashing, gated each release with a plant **production trial run (PTR)**, and reported to **GMs and Directors**.
- Managed **3 suppliers** (infotainment software, Map and camera), including China-based teams, across time zones on builds, release schedules and defect fixes; drove **120+ issues to closure** in JIRA and Azure DevOps.
- Resolved **8+ critical field issues on-site** in Delhi, Mumbai and Hyderabad, debugging customer vehicles from logs and driving **root-cause analysis and corrective fixes** with suppliers.
- Built **CANoe and Vehicle Spy ECU test benches** enabling ICAT certification and EMI-EMC compliance, and took the camera system through **certification testing at ICAT**.

Graduate Engineer Trainee — *Aug 2021 – Aug 2022*

- Validated ADAS, navigation, multimedia, Bluetooth and telematics (e-Call, remote control); automated JIRA bug tracking.

Education & Credentials

University of Nevada, Reno — M.S. Computer Science, GPA 3.9/4.0 — Jan 2025 - Dec 2026

The Maharaja Sayajirao University of Baroda — B.E. Electrical Engineering, CGPA 8.72/10 — Jul 2017 - May 2021

Certifications: Google Project Management Certificate (in progress) • Mastering CAN Network (Vector) • Automotive OTA

Recognition: *Women Engineer of the Year* nominee, Digital Engineering Awards 2023 • *Highest Potential Employee*, MG Motor